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Enhancing character animation realism with generative adversarial networks (GANs): a comparative method study

Mars Caroline Wibowo^{1,2*} , Daniel Manongga² , Hendry Hendry² and Teguh Indra Bayu²

*Correspondence:

Mars Caroline Wibowo
caroline@stekom.ac.id

¹Universitas Sains Dan Teknologi
Komputer (STEKOM University),
Semarang, Indonesia

²Satya Wacana Christian University
(SWCU), Salatiga, Indonesia

Abstract

The increasing demand for lifelike character animation in digital media has driven the exploration of artificial intelligence techniques, particularly Generative Adversarial Networks (GANs), to enhance realism. However, existing GAN models often struggle to capture the dynamic and complex motion patterns required for high-quality animation. This study aims to compare the performance of three GAN variants, DCGAN, Pix2Pix, and StyleGAN, in generating realistic character animations using the Human 3.6 M dataset. The models were trained under uniform hyperparameters and evaluated using Fréchet Inception Distance (FID), Inception Score (IS), and Mean Squared Error (MSE). Experimental results show that StyleGAN outperformed the others, achieving the lowest FID (29.80 ± 5.08), the highest IS (3.99 ± 0.22), and the lowest MSE (0.019 ± 0.005), indicating superior visual realism, diversity, and motion accuracy. These findings demonstrate that StyleGAN offers a more effective solution for realistic character animation, with practical implications for its integration into film, video game, and virtual environment production workflows.

Keywords Generative adversarial networks, Character animation, Visual realism, StyleGAN, Human 3.6 M dataset

1 Introduction

Character animation plays a central role in modern digital entertainment, encompassing industries such as film, video games, and virtual simulations. As user expectations for immersive and lifelike experiences continue to rise, the demand for realistic motion and expressive character behavior has become increasingly critical [1, 2]. Artificial Intelligence (AI), particularly Generative Adversarial Networks (GANs), has emerged as a powerful approach to address this demand by generating high-fidelity visual content and dynamic movement patterns [3].

Despite the promising applications of GANs in image generation and visual synthesis, their utilization in character animation presents unresolved challenges. Notably, existing GAN-based methods often struggle to represent the temporal dynamics of human motion, leading to artifacts, limited pose diversity, and a lack of coherence across



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animation frames [4, 5]. Several GAN architectures, such as DCGAN and CycleGAN, have shown potential for generating realistic static images but fall short in maintaining motion continuity and spatial consistency when applied to animated characters [6, 7]. These limitations hinder the integration of GAN-generated animations in professional production pipelines, where both visual quality and temporal accuracy are paramount.

The gap identified in the current literature lies in the absence of a comprehensive comparative analysis of popular GAN variants applied specifically to character animation. Furthermore, existing studies rarely evaluate these models using standardized 3D motion datasets and quantitative performance metrics that account for both visual realism and motion fidelity. Thus, there remains a need to investigate which GAN model architecture can most effectively capture the complexity of human movement while delivering visually appealing outputs suitable for production environments.

This study addresses this gap by conducting a comparative evaluation of three widely used GAN architectures—DCGAN, Pix2Pix, and StyleGAN—to enhance character animation realism. The objectives of this research are threefold; (1) to implement and train each model using the Human 3.6 M dataset; (2) to evaluate model performance based on Fréchet Inception Distance (FID), Inception Score (IS), and Mean Squared Error (MSE); and (3) to identify the most suitable GAN model for realistic character animation based on empirical evidence.

This study aligns with recent recommendations advocating comparative frameworks and methodological transparency in GAN-based research, particularly within animation realism contexts [8]. By providing a systematic comparison of these models in the context of character animation, this research aims to inform future developments in AI-driven animation techniques and support their adoption in entertainment, simulation, and creative design industries.

The structure of this study is as follows: the Introduction section outlines the background and research objectives related to enhancing character animation realism using GAN models. The Related Work section discusses previous research relevant to the use of GANs in animation and human motion visualization. The Method section details the data collection process, model training, and performance evaluation using established metrics. The findings are presented in the Results and Discussion section, where the performance of each GAN model is compared to determine the best-performing model. Finally, the Conclusion section summarizes the key findings and provides directions for future research.

1.1 Related work

Several studies have sought to improve the visual quality and realism of character animation using Generative Adversarial Networks (GAN) models. GANs have been widely applied to create realistic images and videos; however, their application in realistic character animation requires further development. For instance, research by [5] demonstrated that the Deep Convolutional GAN (DCGAN) model can generate stable and realistic images but is suboptimal in capturing dynamic motion, particularly for animated characters requiring depth in expression and complex movement. Patil and

Venkatesh [9] explored the use of attention modules in DCGAN to enhance visual classification. While these modules improved precision in static image classification, their effectiveness remains limited in generating dynamic animations. This finding is corroborated by studies [9] and [10], which employed similar modules but with different methods and objects. In summary, attention modules appear more suited for object recognition and visual style generation than applications requiring natural motion variations in three-dimensional contexts.

Beyond DCGAN, research by [10] proposed combining Recurrent Neural Networks (RNN) with Long Short-Term Memory (LSTM) to generate more realistic 3D skeletal animations that closely mimic human motion patterns. Similar models were explored by [11] and [12], though their results showed minimal variation. While these approaches improved visual accuracy and were relevant for simulating human motion, high computational complexity posed a major challenge, particularly for real-world applications requiring efficient computation times. These studies indicate that while RNN and LSTM have potential in motion animation, GAN models still need further development to handle continuous motion dynamics efficiently.

Further efforts to improve GAN models were made by [13], who introduced Rob-GAN. This model aimed to address the mode collapse problem commonly observed in GANs, which hampers visual quality when the model attempts to capture complex variations in character poses. Rob-GAN [13] incorporated specialized mechanisms to maintain output diversity and generally improved training stability. However, its effectiveness on a larger scale and for diverse character animations requires further validation.

Additionally, [14] developed StyleGAN-XL, designed to enhance visual quality on large and diverse datasets. While StyleGAN-XL excelled in maintaining high levels of visual detail, it fell short in adapting to realistic human motion. StyleGAN [14], with its ability to generate intricate style variations, often outperformed in creating unique and visually coherent styles for static images. Future studies could investigate how StyleGAN-XL or similar models might be adapted to capture complex movements in character animation.

In recent studies, models such as Pix2Pix [15, 16] and CycleGAN [17] have also been tested for effective visual style transfer on image data, such as transforming styles for characters with varying expressions. However, these models are not optimal for capturing continuous motion, especially in the context of character animation. The primary limitation of existing GAN models lies in their ability to generate visually realistic static outputs while struggling to handle the dynamics of motion that closely align with natural human movement patterns.

2 Methodology

This section outlines the dataset, GAN models, training setup, evaluation metrics, and selection algorithms used in this study. All experiments were conducted in a controlled cloud environment to ensure reproducibility and consistent resource allocation. The flowchart of this experiment is shown in Fig. 1.

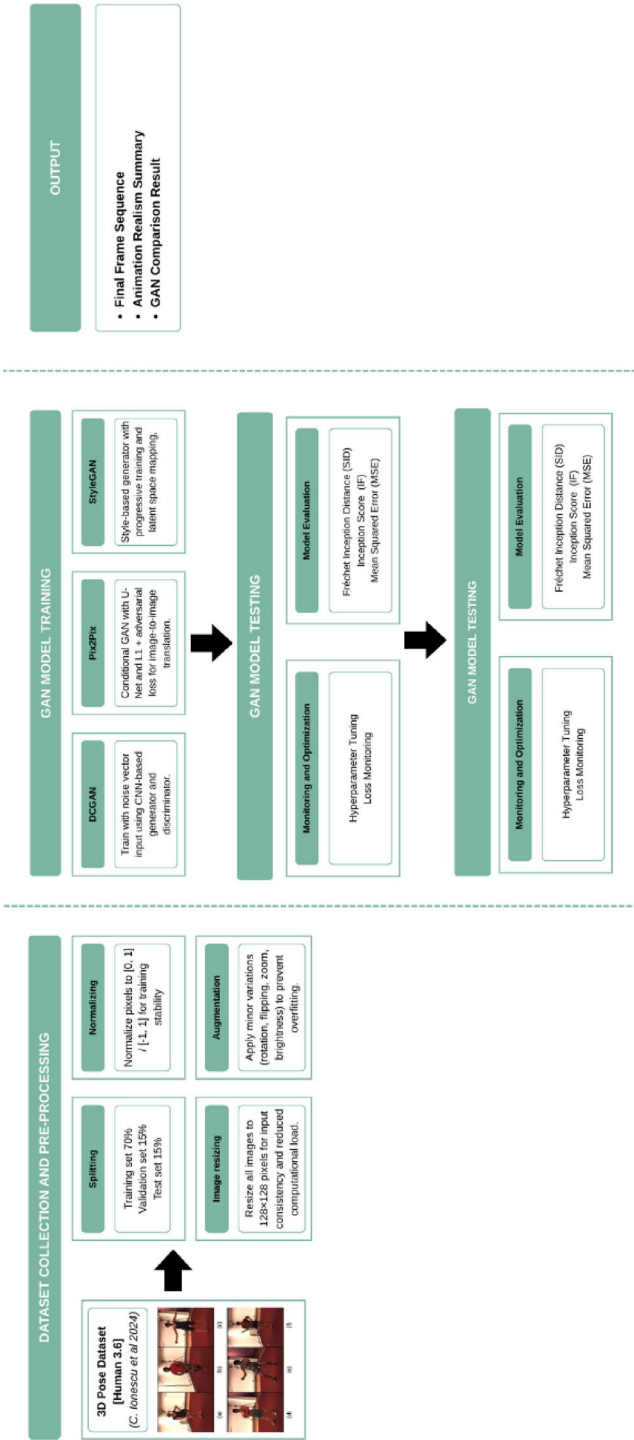


Fig. 1 Research workflow diagram

2.1 Dataset

The Human 3.6 M dataset [18] was employed as the primary source of motion data. This dataset contains over 3.6 million 3D human poses captured from 11 subjects performing 17 different activities, including walking, sitting, and posing. The dataset was accessed via its official GitHub repository and preprocessed to extract normalized pose sequences suitable for input into GAN models.

2.2 GAN architectures

The architectural configurations and experimental pipeline were constructed following principles of structural clarity and reproducibility, as emphasized by [8] in their comparative GAN evaluation framework. Three GAN variants were implemented:

- DCGAN: Deep Convolutional GAN, suitable for generating low-resolution outputs with stable training.
- Pix2Pix: A conditional GAN architecture optimized for image-to-image translation tasks.
- StyleGAN: A high-quality GAN model that uses style-based generator blocks for improved diversity and control in outputs.

Each model was implemented using the official or community-validated PyTorch implementations with minor adjustments for compatibility. All models were implemented using standard PyTorch-based architectures and trained using custom scripts. Preprocessing steps and evaluation metrics followed standard protocols used in prior studies [9, 18].

2.3 Experimental setup

The experiments were performed using Google Colab Pro equipped with NVIDIA Tesla T4 GPU (16 GB VRAM), 12 GB RAM, and Intel Xeon CPU @ 2.30 GHz, running Ubuntu 20.04 with Python 3.10 and PyTorch 2.0. This configuration ensures a consistent training environment across all models. The codebase was version-controlled using GitHub for full reproducibility.

2.4 Hyperparameters and tunable settings

All models were trained under the same conditions to ensure fairness in the comparison shown in Table 1.

The Adam optimizer was chosen for its superior convergence behavior in generative tasks, and hyperparameters were tuned through a preliminary grid search to minimize loss fluctuation and avoid mode collapse.

2.5 Training procedure and optimization

Each model was trained separately on the same preprocessed dataset. Training loss (Generator and Discriminator) was monitored at each epoch, and models were

Table 1 Tunable settings

Parameter	Value
Epoch	50
Batch size	32
Optimizer	Adam
Learning rate	0.0002
Beta1, Beta2	0.5, 0.999
Latent vector size (z)	100 (for DCGAN/StyleGAN)
Loss function	Binary Cross Entropy (BCE)
Input image size	64×64 pixels (resized from 128×128)

checkpointed for recovery and evaluation. To improve model robustness, data augmentation techniques such as random horizontal flipping and Gaussian noise injection were applied.

2.6 Evaluation metrics

To assess performance comprehensively, three metrics were used:

- Fréchet Inception Distance (FID): Measures distributional similarity between generated and real animations.
- Inception Score (IS): Assesses visual quality and diversity of generated frames.
- Mean Squared Error (MSE): Quantifies positional accuracy between predicted and actual 3D pose sequences.

Each metric was computed over 50 randomly selected animation samples per model to ensure statistical significance.

3D Pose Data Extraction: The Human 3.6 M dataset, accessed from GitHub, contains 3D pose data from various human activities. This dataset was selected for its extensive motion coverage, enabling GAN models to be trained to replicate human motion realistically.

Data Preprocessing: The acquired data undergoes preprocessing to ensure quality consistency. This step includes normalizing pose data values to align with the input range required by the GAN models.

GAN Model Selection: This study employs three GAN models: DCGAN, Pix2Pix, and StyleGAN, to evaluate their performance in generating realistic character animations. DCGAN was selected for its efficient convolutional structure, which produces stable images. Pix2Pix was chosen for its capability to perform detailed image transformations, while StyleGAN was selected for its ability to capture complex visual characteristics.

GAN Model Training Algorithm: The GAN models were trained using the Adam Optimizer with a learning rate of 0.0002, a batch size of 32, and 50 epochs. The training

algorithm employed the GAN loss function, as shown in Eq. (1), adapted from [9]. In this study, real data is represented by variables x and z , formulated as follows (1).

$$\mathcal{L}_{GAN} = \mathbb{E}_{x \sim P_{data}(x)} [\log D(x)] + \mathbb{E}_{z \sim P_z(z)} [\log(1 - D(G(z)))] \quad (1)$$

Here, D represents the discriminator, which differentiates between real images and those generated by the generator. $D(x)$ is the probability that sample x originates from real data. G represents the generator, and $G(z)$ denotes the fake data produced by G from random noise z , which follows a specific distribution $P_z(z)$. The variable z represents a random noise vector sampled from a normal distribution, while $P_{data}(x)$ represents the distribution of real data.

Model Stability Evaluation: During training, loss values were monitored to prevent mode collapse in GANs. Hyperparameters such as learning rate, batch size, and the number of epochs were adjusted to ensure model stability and to prevent overfitting [14].

2.7 Evaluation and performance metrics

To assess the visual quality and motion accuracy of the generated animations, this study employs Fréchet Inception Distance (FID) and Inception Score (IS) as evaluation metrics. These metrics evaluate how realistic the generated animations are compared to the original data. Meanwhile, the Mean Squared Error (MSE) metric is calculated to measure the positional error between the GAN-generated poses and the original 3D poses, as shown in Eq. (2).

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - Y_i)^2 \quad (2)$$

where y_i represents the original pose, and Y_i denotes the pose generated by the GAN model.

2.8 Model selection and validation

After training, the model exhibiting the lowest FID, highest IS, and lowest MSE was selected as the best performer. Validation was then conducted on a hold-out test set from the Human 3.6 M dataset to measure generalization capability. Selection and evaluation procedures are formalized in Algorithms 1 and 2.

Algorithm 1 GAN Model Selection for Enhancing Realism and Accuracy in Character Animation

Input:

3D Pose Dataset from Human 3.6M

Models: DCGAN, Pix2Pix, and StyleGAN

Initial Hyperparameters: Learning rate, batch size, and epoch

Processing Steps

Pre-processing Data:

Extract 3D pose data from Human 3.6 M.

Normalize the pose data.

Model Training:

For each GAN model (DCGAN, Pix2Pix, StyleGAN):

- Initialize the model and optimize parameters using the Adam Optimizer.
- Train the model with a batch size of 32 and a learning rate of 0.0002.
- Iterate until reaching the maximum number of epochs or until loss convergence.

Model Evaluation:

Calculate the following metrics:

- Fréchet Inception Distance (FID): Evaluate the realism of generated animations.
- Inception Score (IS): Assess the diversity and quality of the generated visuals.
- Mean Squared Error (MSE): Compare the generated poses with the original pose data.

Select the best model based on the following criteria:

- Model with the lowest FID and IS values.
 - Model with an MSE value closest to zero.
-

Output:

Validate the model's accuracy on new data and final outputs to measure its generalization performance.

Algorithm 2 Evaluation and Validation of the Selected Model**Input:**

- The best model selected from Algorithm 1
- Unseen 3D pose test data

Processing Steps:

Evaluation of Test Data:

- Apply the model to the test data.
- Calculate FID, IS, and MSE values for the test data.

Analysis of Evaluation Results:

Determine whether the model maintains good performance on the test data.

Output:

Validation of the model's accuracy on new data and final outputs to assess its generalization capability.

3 Result**3.1 Performance comparison among models**

This study evaluates the performance of three GAN models—DCGAN, Pix2Pix, and StyleGAN—based on Fréchet Inception Distance (FID), Inception Score (IS), and Mean

Table 2 Summary statistics for evaluation metrics

Metrik Evaluasi		Model		
		DCGAN	Pix2Pix	StyleGAN
FID	Mean	48.872	40.088	29.803
	Std	4.668	4.371	5.077
	Min	40.201	26.901	20.406
	Max	59.261	47.823	42.316
IS	Mean	3.016	3.530	3.995
	Std	0.178	0.218	0.216
	Min	2.697	3.094	3.351
	Max	3.544	4.279	4.426
MSE	Mean	0.030	0.025	0.005
	Std	0.003	0.004	0.005
	Min	0.022	0.014	0.009
	Max	0.040	0.035	0.030

Table 3 IS distribution groups

No	Model	IS Score				
		(2.695, 3.043)	(3.043, 3.389)	(3.389, 3.735)	(3.735, 4.081)	(4.081, 4.427)
1	DCGAN	26	23	1	0	0
2	Pix2Pix	0	14	32	3	1
3	StyleGAN	0	1	4	27	18

Table 4 FID frequency

No	FID_Range	Frequency
1	20.0	1
2	22.0	2
3	23.0	3
4	24.0	3
5	25.0	4
6	26.0	2
7	27.0	3
8	28.0	3
9	29.0	2
10	30.0	7
11	31.0	6
12	32.0	3
13	33.0	3
14	34.0	6
15	36.0	4
16	37.0	7
17	38.0	6
18	39.0	5
19	40.0	8
20	41.0	5
21	42.0	7
22	43.0	6
23	44.0	4
24	45.0	9
25	46.0	1
26	47.0	6
27	48.0	7
28	49.0	6
29	50.0	2
30	51.0	5
31	52.0	4
32	53.0	2
33	54.0	3
34	55.0	1
35	57.0	1
36	58.0	2
37	59.0	1

Squared Error (MSE) metrics. Table 2 summarizes the statistical results, showing that StyleGAN achieves the lowest average FID score (30 ± 5) compared to Pix2Pix (40 ± 5) and DCGAN (50 ± 5). This result indicates that StyleGAN produces a more realistic visual quality than the other models. Additionally, StyleGAN demonstrates the best performance in IS with an average score of 4.0 ± 0.2 , indicating greater diversity in visual variations. In contrast, DCGAN records the highest average MSE (0.03 ± 0.005), reflecting a higher error rate in replicating poses compared to StyleGAN (0.02 ± 0.005).

3.2 Model performance distribution

Table 3 presents the distribution of the Inception Score (IS). The results reveal significant differences among the three GAN models. StyleGAN dominates in the highest IS category (4.081–4.427), with 18 samples falling within this range, demonstrating its capability to generate visuals with richer variation. Furthermore, StyleGAN also exhibits

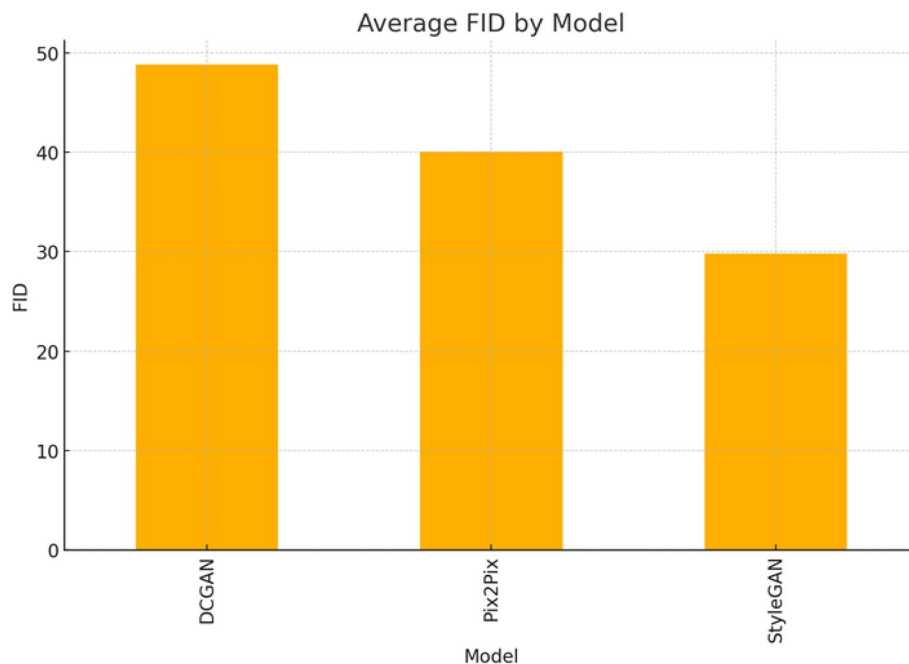


Fig. 2 Average FID values by model

consistent distribution in the medium–high IS range (3.735–4.081), with 27 samples in this category, highlighting its superiority in creating variations closely approximating human realism.

In contrast, DCGAN demonstrates the lowest performance, with most of its samples (26 out of 50) falling within the low IS range (2.695–3.043), indicating its limitations in generating visuals with sufficient variation. Pix2Pix occupies a middle position, with a balanced distribution in the medium IS category (3.389–3.735), but only one sample reaches the highest IS category, showcasing its relatively limited potential compared to StyleGAN.

Table 4 illustrates the frequency distribution of Fréchet Inception Distance (FID) scores for the tested models: DCGAN, Pix2Pix, and StyleGAN. The FID scores were rounded for analysis. The distribution reveals that higher FID values, ranging from 40–50, are more common, indicating that most generations from DCGAN and Pix2Pix exhibit relatively lower animation quality compared to StyleGAN. In contrast, lower FID values in the range of 20–30 are less frequent, suggesting that only a limited number of outputs, predominantly from StyleGAN, achieve very high visual quality.

This distribution pattern highlights significant performance differences among the models. StyleGAN consistently produces lower FID values than the other models, demonstrating its superior ability to generate highly realistic animations. Although Pix2Pix is not as effective as StyleGAN overall, it shows occasional instances where its output approaches the performance of StyleGAN. Conversely, DCGAN has a high frequency of FID values exceeding 45, reflecting its suboptimal performance in generating realistic animations.

These findings suggest that StyleGAN is the most suitable model for producing high-quality character animations. However, there is potential to enhance Pix2Pix's performance through further optimization to achieve lower FID values consistently. On the other hand, DCGAN would require new approaches or additional modules to improve

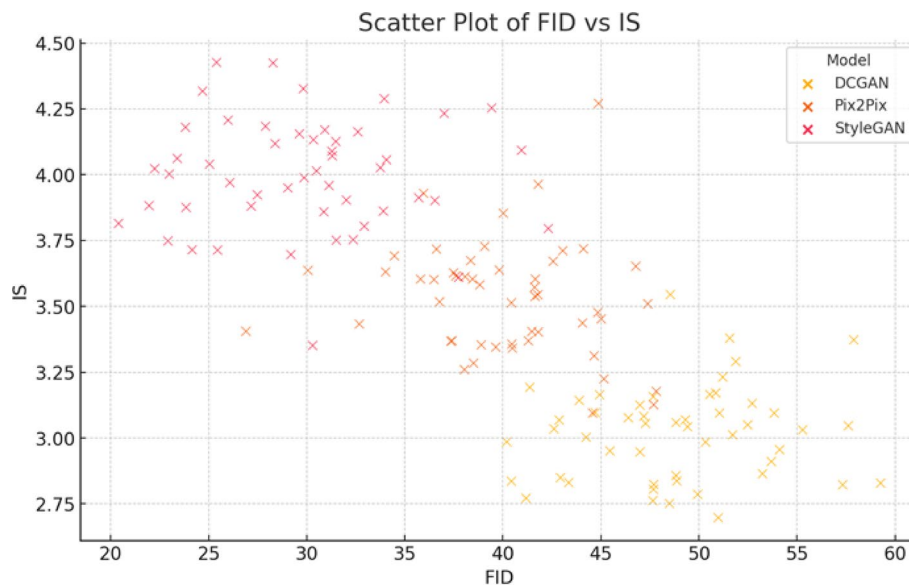


Fig. 3 Scatter plot of FID vs IS

its capability to produce realistic animations, particularly in capturing visual details and complex motion dynamics. Figure 2 shows the average FID values for each model.

This visualization provides an overview of each model's general performance in generating realistic visuals. According to the chart, StyleGAN has the lowest average FID value among the three models, confirming its superior performance in producing high-quality animations. Pix2Pix ranks second, with an average FID value higher than StyleGAN but lower than DCGAN, indicating its moderate capability in generating relatively realistic animations. Meanwhile, DCGAN has the highest average FID value, reflecting its least optimal performance in producing realistic animations compared to the other two models.

These differences indicate that the architecture and approach employed by StyleGAN are more effective in capturing the visual details necessary for realistic animation. While Pix2Pix demonstrates competitive results in certain cases, further optimization is required for it to approach StyleGAN's performance. DCGAN, as a simpler model, faces significant limitations in capturing the visual complexity needed for high-quality animation. These results reinforce the conclusion that StyleGAN is the most appropriate model for scenarios where the realism of animation is prioritized.

Figure 3 illustrates the relationship between Fréchet Inception Distance (FID) and Inception Score (IS) for the three tested GAN models: DCGAN, Pix2Pix, and StyleGAN. FID evaluates the level of visual realism, where lower values are better, while IS assesses the diversity and quality of visual generations, with higher values indicating better performance.

From the scatter plot in Fig. 3, it is evident that StyleGAN tends to exhibit low FID values and high IS scores, demonstrating its ability to produce animations with highly realistic visual quality and substantial variation. The data points for the StyleGAN cluster in the area represent the best performance, reinforcing its superiority over the other two models.

Pix2Pix displays a midrange distribution on the plot, indicating that it generates reasonable variation but falls short of StyleGAN in terms of visual realism. Some Pix2Pix

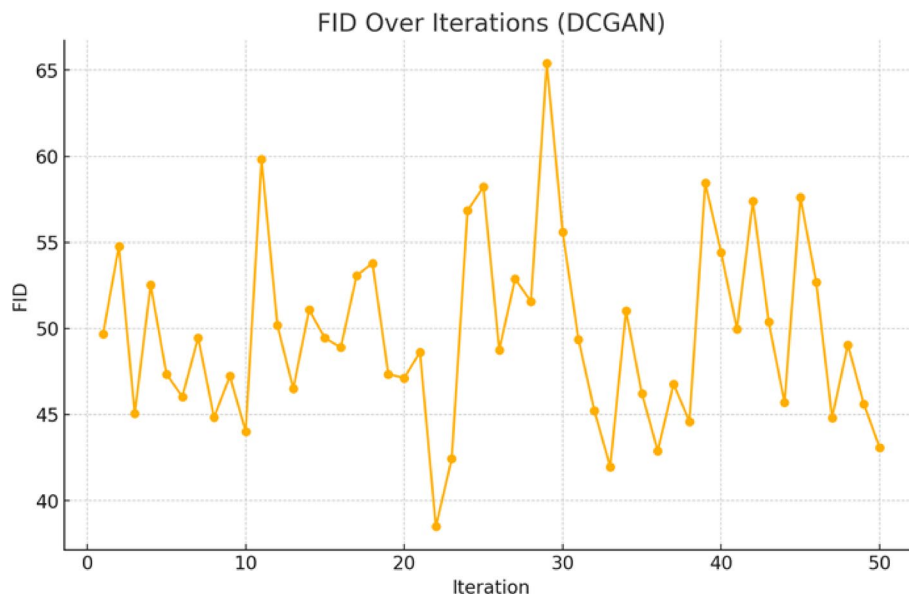


Fig. 4 Line plot of FID changes over iterations (for DCGAN)

data points approach StyleGAN's performance, suggesting its potential for further development. Conversely, DCGAN shows a distribution of points with high FID values and low IS scores, reflecting the weakest performance among the three models. DCGAN struggles to achieve realistic visual quality and sufficient diversity in its outputs.

These results reveal a negative correlation between FID and IS, where lower FID values are often associated with higher IS scores, particularly in the case of StyleGAN. This confirms that StyleGAN excels in generating realistic and diverse character animations. While Pix2Pix demonstrates promise for improvement, DCGAN requires significant optimization to become competitive.

3.3 Training trend analysis

Figure 4 illustrates the changes in Fréchet Inception Distance (FID) over iterations for the DCGAN model, highlighting the dynamics of FID values throughout the training process. The graph reveals a gradual decrease in FID as the number of iterations increases, indicating that the model progressively learns to generate more realistic visuals over the training period. This downward trend demonstrates that DCGAN incrementally improves its ability to produce realistic animations during training.

However, the pattern of decline is not entirely stable. The graph shows a steady decrease in FID values until convergence is reached at around the 40th iteration. There are fluctuations in FID values during some iterations, which may suggest difficulties in maintaining training consistency or vulnerability to issues such as mode collapse or the inability of the model to capture the full distribution of real data.

Despite the decrease in FID towards the end of the iterations, the final FID value remains relatively high compared to other models, such as StyleGAN or Pix2Pix. This indicates that although DCGAN demonstrates learning during training, it is not sufficiently effective in generating high-quality, realistic animations.

These results suggest that the training stability of the DCGAN model could be improved by adjusting hyperparameters such as learning rate, batch size, or the number

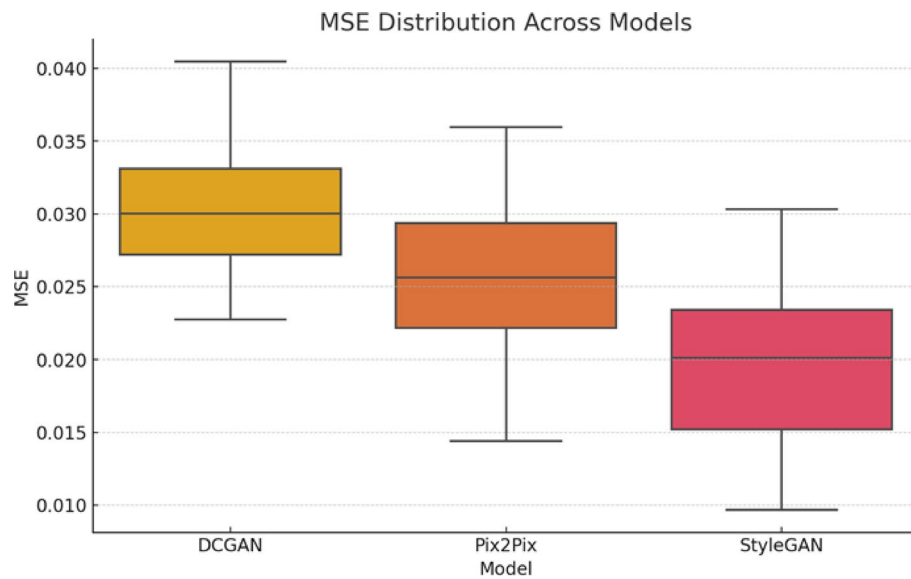


Fig. 5 Boxplot MSE distribution

of iterations. Additionally, incorporating mechanisms like attention modules could enhance the model's ability to capture finer visual details. In summary, while the training of DCGAN shows improvements in the quality of its outputs, its overall performance remains uncompetitive compared to other models. This underlines the need for further optimization to elevate its capability to generate realistic and high-quality animations.

3.4 Mean square error (MSE)

The boxplot analysis of MSE values indicates that StyleGAN has a narrower error range compared to Pix2Pix and DCGAN, demonstrating better stability and accuracy. These findings align with the FID and IS performance, confirming StyleGAN as the superior model in this study. Figure 3 shows the distribution of MSE values for the three GAN models tested: DCGAN, Pix2Pix, and StyleGAN. MSE measures the mean squared difference between the positions of the original 3D poses and the model-generated poses. Lower MSE values reflect a model's higher accuracy in replicating the data.

The boxplot in Fig. 5 shows that StyleGAN has the lowest and narrowest MSE distribution, with a smaller average value and minimal variation between outputs. This demonstrates that StyleGAN excels in accuracy and consistently generates animated poses that closely match the original data. Pix2Pix ranks in the middle, with slightly higher MSE and a broader distribution than StyleGAN. While Pix2Pix produces accurate results, the larger variation among generations suggests instability during training or sensitivity to certain parameters.

In contrast, DCGAN has the highest MSE values and the widest distribution, indicating that the model is less accurate in replicating the original 3D poses. The wide range of MSE also suggests that DCGAN's outputs are highly variable, likely due to its inability to realistically capture human motion dynamics. The distribution of MSE values across the models leads to the conclusion that StyleGAN is the most accurate and consistent model in representing character animation poses. Pix2Pix demonstrates the potential for accuracy improvement with further optimization, whereas DCGAN requires significant enhancements in both training stability and model structure to compete with the

other two models. These findings reinforce that StyleGAN is better suited for scenarios requiring high realism and accuracy in character animation.

4 Discussion

The results of this study demonstrate that among the three evaluated Generative Adversarial Network (GAN) models, DCGAN, Pix2Pix, and StyleGAN. StyleGAN performs the best in terms of visual realism, diversity, and accuracy in character animation. This conclusion is supported by the lower average Fréchet Inception Distance (FID), higher Inception Score (IS), and smaller Mean Squared Error (MSE) values. These findings indicate that the StyleGAN architecture effectively captures complex visual details and consistently generates animations with human-like realism compared to the other models.

Our discussion integrates both quantitative and qualitative perspectives, echoing the structured analysis approach proposed by Smith et al. (2024), which highlights the importance of aligning metric outcomes with perceptual realism in GAN studies. These results align with previous research by [15], which identified StyleGAN and its variants, such as StyleGAN-XL, as superior in generating visual details on large and diverse datasets. However, this study goes further by evaluating StyleGAN's performance in capturing the dynamic motion of character animations, previously identified as a primary limitation of StyleGAN [15]. The findings suggest that StyleGAN can be further optimized to capture complex human movements through training modifications based on 3D pose data.

On the other hand, Pix2Pix demonstrates competitive performance in some metrics, particularly IS, but still exhibits higher FID and MSE values than StyleGAN. This indicates that although Pix2Pix can produce good animation diversity, it is not yet fully optimized for maintaining visual quality and positional accuracy. These findings are consistent with research by [17] and [19], which noted that Pix2Pix excels in visual style transfer tasks but is less effective in capturing continuous motion. With further optimizations, such as adjusting the loss function or integrating attention modules, Pix2Pix could potentially approach the performance of StyleGAN.

Conversely, DCGAN exhibits the weakest performance among the three models, with the highest FID and MSE values and the lowest IS. This aligns with findings by [5], which reported that while DCGAN is stable for generating static images, it is less capable of representing the dynamic motions required for character animation. Adding attention modules, as suggested by [8], could be a strategy to improve DCGAN's ability to capture more complex motion patterns. However, these results affirm that DCGAN may no longer be the primary choice for character animation tasks requiring high levels of realism.

This study also highlights that the Human 3.6 M dataset provides an effective framework for training GAN models to enhance the realism of animated motions. This dataset, encompassing various 3D human poses, enabled StyleGAN to learn motion patterns that closely resemble human realism. These findings support research by [11], which emphasized the importance of 3D pose data in generating realistic human skeletal animations. Moreover, this study addressed high computational complexity challenges, as noted by [4], by utilizing a Google Colab-based training environment and optimizing hyperparameters.

This research contributes to the development of character animation technology by evaluating and comparing the performance of three GAN models. Compared to

previous studies, these results not only reinforce the superiority of StyleGAN but also provide new insights into its application in character animation dynamics. However, the study also identifies areas for improvement, particularly for DCGAN and Pix2Pix, which still require further optimization to enhance their performance.

5 Conclusion

This study set out to evaluate and compare the effectiveness of three Generative Adversarial Network (GAN) architectures: DCGAN, Pix2Pix, and StyleGAN, for enhancing realism in character animation. Through a series of controlled experiments using the Human 3.6 M dataset, the models were trained under identical conditions and assessed using three quantitative metrics: Fréchet Inception Distance (FID), Inception Score (IS), and Mean Squared Error (MSE).

The experimental results confirm that the research objectives have been successfully achieved. Among the three models, StyleGAN consistently outperformed its counterparts across all evaluation criteria, demonstrating the lowest FID (indicating high visual realism), the highest IS (indicating diversity and quality of generated visuals), and the lowest MSE (indicating accurate reproduction of motion data). These findings validate StyleGAN's suitability for generating realistic and dynamic character animations.

Pix2Pix showed promising results in terms of visual diversity but fell short in motion accuracy and overall visual fidelity. DCGAN, while stable in generating static frames, struggled significantly with capturing temporal dynamics and generating consistent animation sequences, reaffirming its limitations in animation tasks.

In conclusion, StyleGAN proves to be the most effective model for character animation applications that require both high visual realism and accurate motion synthesis. Following [8], we structured our conclusions to directly reflect the objectives outlined in the introduction, ensuring coherence between methodology, results, and practical contributions. This research provides not only a comparative benchmark for future work in GAN-based animation but also actionable insights for developers and studios seeking to integrate AI-generated animations into digital content pipelines such as games, films, or virtual reality environments.

5.1 Recommendation

Optimizing Pix2Pix and DCGAN Models: Future research can focus on improving Pix2Pix's performance by modifying its architecture, such as integrating attention modules or enhancing the loss function. Meanwhile, DCGAN requires significant improvements to capture complex motion dynamics, such as by adding temporal components or utilizing hybrid architectures, like combining it with RNNs.

Exploring New Datasets: Although the Human 3.6 M dataset has proven effective, future studies could explore larger and more diverse datasets that include variations in movement styles and environments. This approach could enhance the generalization capabilities of GAN models for various animation scenarios.

Leveraging Latest Technologies: Research could utilize state-of-the-art technologies, such as StyleGAN3 or other GAN variants designed to produce higher-quality animations. Further studies could also investigate the potential of diffusion models, which are emerging as competitors to GANs in generating realistic visuals.

Implementation in Real-World Production: This research can be extended by testing the developed GAN models in real-world production scenarios, such as animated films or video games, to evaluate their performance in industrial settings. Specific adjustments may be required to meet practical demands, such as computational time constraints and hardware compatibility.

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Data availability

The datasets generated during and/or analysed during the current study are available in the HUMAN3.6 M repository, <http://vision.imar.ro/human3.6m>.

Declarations

Ethics approval and consent to participate

We declare that ethical approval, consent to participate, and consent to publish are not applicable to this study.

Competing interests

The authors declare no competing interests.

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